Practice Quiz: Learning

Part A: Multiple Choice

1. A beginning crocheter improves rapidly in their first hour of practice, then improves more slowly over the following weeks. This pattern most closely resembles: A) A neural network with a constant learning rate B) A neural network with a high initial learning rate that decays over time C) A neural network that has overfit to its training data D) A neural network that has not been given any training data

2. A crocheter who has spent years making only granny squares executes them perfectly but struggles with unfamiliar patterns. In machine learning terms, this crocheter has: A) Generalized well across all crochet tasks B) Overfit to a single pattern type, limiting their ability to handle new tasks C) Achieved the global minimum of their loss landscape D) Regularized their training through diverse practice

3. Which of the following crochet practices best functions as **regularization** (preventing overfitting)? A) Making the same pattern repeatedly until it is perfect B) Always using the same yarn weight and hook size C) Deliberately trying new stitch types, yarn weights, and project styles D) Avoiding all challenging patterns

4. The loss landscape of a simple single-crochet dishcloth compared to a complex lace shawl is: A) More rugged with many local minima B) Smoother and more forgiving — fewer ways to go seriously wrong C) Identical in complexity D) Impossible to compare

5. When a crochet stitch becomes "muscle memory," the crocheter can execute it without conscious attention. In neural network terms, this means: A) The weights have been deleted B) The network has overfit C) The weights have converged — the motor policy produces consistent, accurate output automatically D) The network has stopped training entirely

6. A crocheter who switches from smooth acrylic yarn to textured handspun wool and struggles with tension is experiencing: A) Overfitting — their skills are too general B)

Distribution shift — the new input (yarn type) differs from their training data C) A decrease in learning rate D) An improvement in generalization

7. A crochet pattern book that progresses from dishcloths to scarves to hats to sweaters is structured like: A) Random sampling from the full task distribution B) Curriculum learning — training on progressively harder examples C) Dropout regularization D) Adversarial training

Part B: Short Answer

1. A beginning crocheter attempts front post double crochet for the first time. Over 5 rows of practice, their error rate drops from many mistakes per row to nearly zero. Sketch a plausible **training curve** (errors on the vertical axis, rows on the horizontal axis) and explain where the steepest improvement occurs and why.

2. Explain the overfitting/generalization tradeoff using a specific crochet example. Describe a crocheter who has overfit, explain what their skill looks like in practice, and propose a "training regimen" (new projects, techniques, or materials) that would improve their generalization.

3. A complex lace pattern has a "rugged loss landscape" — many ways to go wrong and narrow paths to success. A simple scarf pattern has a "smooth loss landscape" — hard to go wrong. Using the active inference framework, explain how the crocheter's generative model differs between these two projects and how the size of prediction errors relates to the landscape's ruggedness.